



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

System Documentation (SD)

Koperasi KADA All-in-one Financial Platform

Version 1.0

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Faculty of Computing

Prepared by: Tech EcoPioneer

Revision Page

a. Overview

Describe the content of the current version of SD (see below – the note in red).

b. Target Audience

State the targeted audience for the SD.

c. Project Team Members

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d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Evelyn Goh Yuan Qi		
...			

Note:

This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly

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1. Introduction

This Software Documentation (SD) provides a detailed overview of the Koperasi KADA Online Platform System. The document focuses on the Software Requirements Specification (SRS), which explains what the system should do and how it should work. This includes the functional and non-functional requirements for developing and deploying the system.

The SD will help everyone involved in the project to have a clear understanding of what is needed to build the system and the expectations of the result system. It acts as a guide for the developers, testers, and other stakeholders.

1.1 Purpose

The goal of this SD is to document all the requirements of the Koperasi KADA Online Platform System. It ensures alignment amongst stakeholders and provides a basis for the development process. This helps to ensure that the system meets user needs and achieves project objectives.

The primary audience for this SD includes software developers who will be implementing the system according to the specified requirements, project manager who will be ensuring that the development aligns with the defined goals and requirements, as well as the client and stakeholders who will be verifying the requirements reflected in the system for their needs and requirements.

1.2 Scope

The software product the Koperasi KADA Online Platform System is an integrated cloud-based financial management solution that could be used for streamlining or digitizing cooperative operations. This platform will enable members to register, manage savings accounts, apply for loans, and set recurring savings payments through a user-friendly interface. Staff will be able to validate documents, approve applications for registration, process loans, and manage customer support tickets. The directors shall have the capability to access financial reports, approve loans, and monitor system performance. Other significant features include OTP-based login through WhatsApp, automated notifications via email, transaction history, and a user assistance chatbot. It will not support non-financial cooperative operations like inventory or retail management. Its main concentration will be financial and member-related functions with a view to optimizing workflow and enhancing user experience within cooperatives. This platform is especially fitted for the application to

iron out some inefficiencies from the old manual cooperative system by automating critical processes that reduce administrative workload and improve security. Among its objectives are financial inclusivity, transparency, and user satisfaction through digital tools in an accessible and efficient manner. The major benefits derived from it are smooth operation efficiency, reduced errors, increasing member engagement, and scalability for the growth of the cooperative. This specification meets higher-level requirements and system objectives, providing a secure, scalable, and user-friendly solution for cooperative financial management.

1.3 Definitions, Acronyms and Abbreviation

In order to ensure clarity and consistency throughout this document, the following definitions, acronyms and abbreviations are used.

1. **SRS:** Software Requirements Specification
2. **Feedback System:** a software solution used to collect, analyze and report students' feedback.
3. **Board of Director:** A group of people that provides expertise for a company or organization
4. **SD:** Software Documentation
5. **OTP (One-Time Password):** A secure authentication method where a unique password is generated for a single login session or transaction.
6. **ERD (Entity-Relationship Diagram):** A visual representation of the data and relationships in a system.
7. **KADA (Lembaga Kemajuan Pertanian Kemubu):** A cooperative institution focusing on agricultural and financial management.
8. **API (Application Programming Interface):** A set of rules and protocols for building and interacting with software applications.
9. **PDPA (Personal Data Protection Act 2010 (Malaysia)):** A legal framework for data privacy and protection in Malaysia.
10. **UAT (User Acceptance Testing):** The process where end-users validate the functionality and performance of a system before its official release.
11. **KPI (Key Performance Indicator):** Metrics used to evaluate the success and performance of specific objectives.
12. **Scalability:** The ability of the system to handle increasing user demands or workload.
13. **Concurrent Users:** The number of users accessing the system simultaneously.

1.4 References

- 1) *Portal Lembaga Kemajuan Pertanian Kemubu | KADA KEMUBU*. (n.d.). <http://www.kada.gov.my/>
- 2) *CyberSecurity Malaysia*. (n.d.). https://www.cybersecurity.my/en/knowledge_banks/principles_guidelines/main/detail/2339/index.html
- 3) Hunt, J. (2000). *An introduction to the UML and the unified process*. In *Practitioner series* (pp. 19–33). https://doi.org/10.1007/978-1-4471-3639-2_3
- 4) Lane, G. K. (2023, January 17). *How to write an SRS Document (Software Requirements Specification Document)*. Perforce Software. <https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document>
- 5) *Software Testing Documentation: Types & Examples | PFLB*. (2019, November 5). PFLB. <https://pflb.us/blog/importance-of-software-testing-documentation/>

1.5 Overview

The System Documentation (SD) for the All-in-One Financial Platform for KADA Cooperative provides a comprehensive outline of the key aspects of the software development lifecycle, ensuring alignment with the project's objectives and requirements. This document is structured to guide stakeholders, including developers, testers, and end-users, by detailing the system's purpose, design, and implementation. It begins with an introduction that defines the purpose, scope, and references of the project, followed by specific requirements, including personas, system features, and use cases that describe the system's expected functionality. Besides, the User Story Details section further breaks down individual user stories, presenting their descriptions, flow of events and acceptance criteria, supported by diagrams like sequence and activity diagrams for visual clarity. Non-functional requirements, such as usability, reliability, and performance, are addressed to ensure the system meets quality standards. Additionally, the document outlines any design constraints, such as hardware and software limitations, imposed by the operational environment. Overall, the SD provides a comprehensive, structured approach to ensure traceability, collaboration, and successful implementation throughout the project lifecycle.

2. Specific Requirements

*This section of SD serves as the SRS that should contain all the software requirements to a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, operators, or other external systems. These requirements should **include at a minimum a description of every input (stimulus) into the system, every output (response) from the system, and all functions performed by the system** in response to an input or in support of an output.*

2.1 Persona

2.1.1 Persona 1 (New User)

User Need

As a new user, an employee of KADA needs a seamless way to register as a cooperative member by providing their personal and employment details. The system must enable them to complete the registration process efficiently and securely, ensuring they can gain access to cooperative benefits, such as savings accounts and loan applications, once their membership is approved.

User Stories

1. As an employee of KADA, I want to register as a cooperative member by providing the necessary personal details and documents so that I can gain access to the cooperative's financial services.
2. As an employee of KADA, I want to receive a notification once my registration is approved so that I can begin accessing the system's features.

2.1.2 Persona 2 (Member)

User Need

A **member of KADA** needs a way to manage their savings by creating an automated recurring payment, which is an online saving system. Besides that, a way to apply for loan and track application status is required so that it increases the efficiency of financial resources accessibility.

User Stories:

- 1) As a member of KADA, I want to create and manage savings accounts with automated recurring payments, so that I can save money and monitor my financial progress.
- 2) As a member of KADA, I want to apply for loans and track the status of my applications, so that I can access financial resources conveniently and efficiently.

2.1.3 Persona 3 (Staff(clerk))

User Need

Clerks need a streamlined way to approve or reject applications, validate documents, address member inquiries, and generate reports to maintain efficient operations and support management decision-making.

User Stories

1. As a clerk, I want to approve or reject member registrations so that only eligible individuals can access the cooperative's services.
2. As a clerk, I want to validate documents submitted by members so that all applications are accurate and complete.
3. As a clerk, I want to respond to support tickets so that members' inquiries and issues are resolved quickly and efficiently.
4. As a clerk, I want to generate detailed reports for management so that the cooperative's performance and activities are well-documented and easily reviewed.

2.1.4 Persona 4 (Board of Directors)

User Need

As one of the members in the Board of Directors, I need a system that allows me to access detailed financial reports, review loan applications, and monitor system performance. This will help me make informed decisions, approve large loans, and oversee the corporation's overall efficiency.

User Stories

1. As a director, I want to view comprehensive financial reports so that I can evaluate the corporation's financial performance.
2. As a director, I want to review and approve large loan applications to ensure the cooperative operates within financial limits.
3. As a director, I want access to a dashboard that summarises the key metrics and activities to help me make informed decisions.

2.2 System Features

*Specify the product perspective.
Example.*

The Emotion Chatbot Analyzer is a software system that operates on mobile devices, including both Android and iOS operating systems. The system provides a means for UTM students to better manage their emotional health and at the same time simplifies the process of seeking emotional support from UTM counselors. The system also implemented a real-time cloud-based database for better user experiences.

Specify the product module and function..

Include use case diagram here – see an example of Emotion Chatbot Analyzer below

Then, provide the description of each module and function using a table.

Example.

The system features is illustrated in Figure X.X below. The detail description of each module and functions is tabulated in Table X.X.

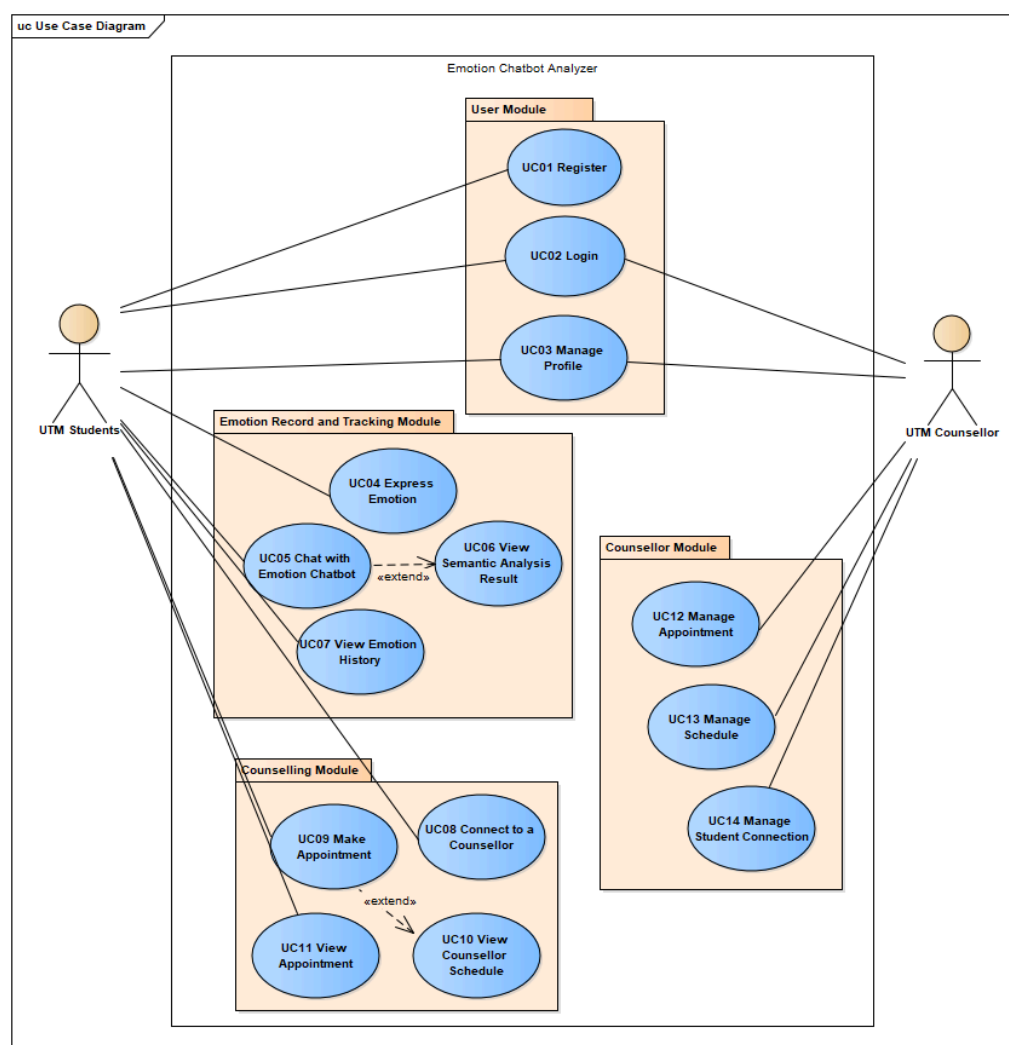


Figure X.X: Use Case Diagram for <Name of the System>

Table X.X: Description of Module and Functions for <Name of the System>

Use case	Function	Description
UC001	Register	This use case allows students to sign up as a user for the mobile application.

[Include domain model i.e. **class diagram** without the operation part – only attributes without details on visibility and type, explain each class and its attributes including the relationships among the classes, see the example.]

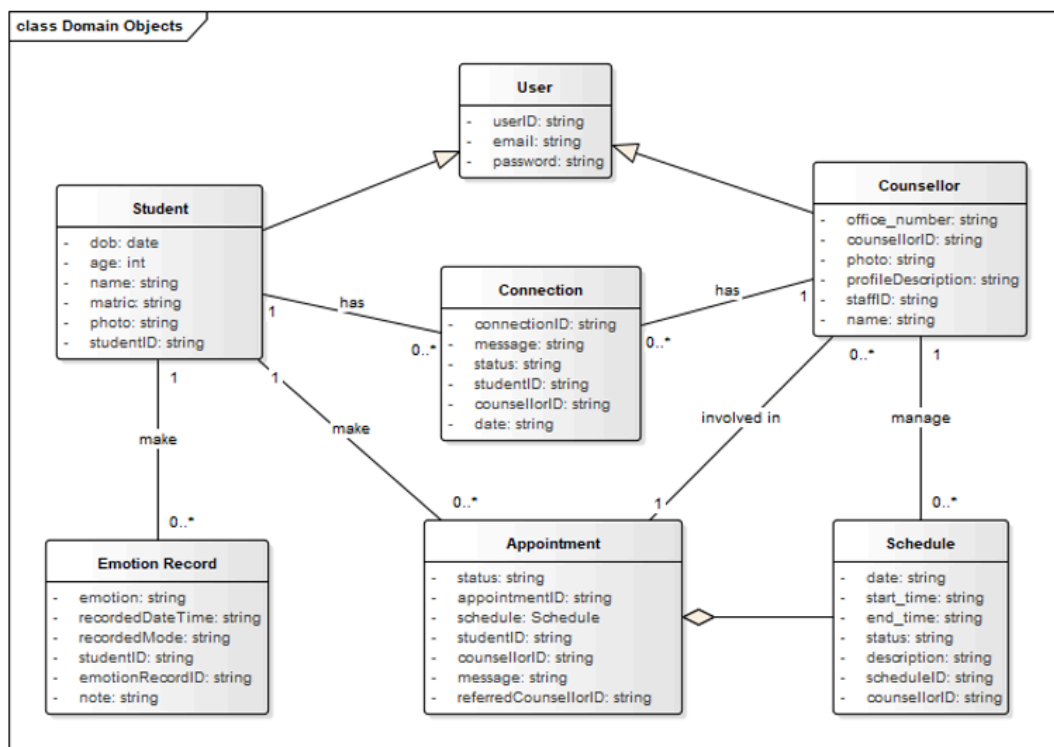


Figure X.X: Domain Model for <Name of the System>

[Include domain model i.e. **class diagram** without the operation part – only attributes without details on visibility and type, explain each class and its attributes including the relationships among the classes, see the example. For the **class with the states only**, consider including its state machine diagram, see the example for state machine diagram of Account class]

Figure 1.3 shows the state diagram for the member application process. After the member application has been submitted, the staff will check and validate the applications. When the application is approved, new users will be updated by receiving email.

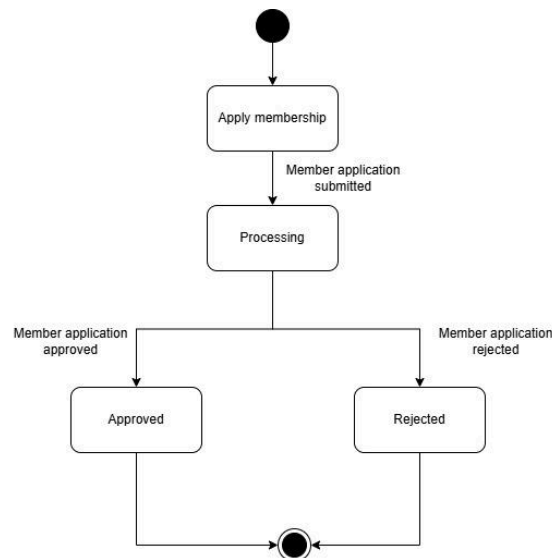


Figure 1.3: State Machine Diagram for Member Application

Figure 1.4 shows the state diagram for the loan application process. The loan application can only be applied by the members of Koperasi KADA. After the loan application has been submitted, it will be checked and validated by the board of directors. After validated by the board of directors, the status of the loan application will be updated to the member.

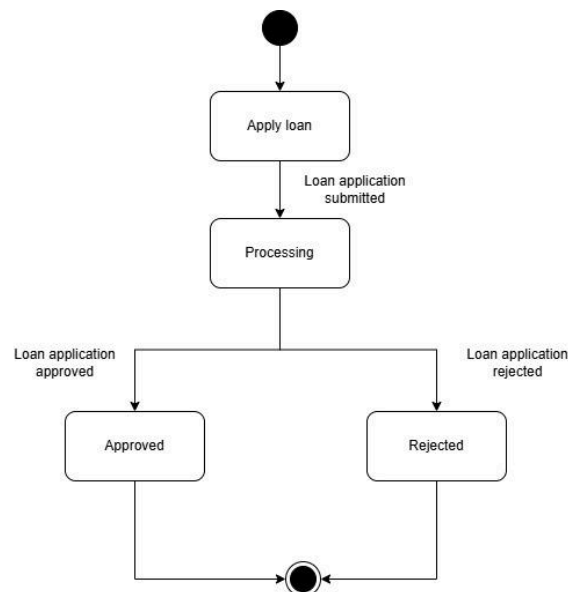


Figure 1.4: State Machine Diagram for Loan Application

2.3 Launch Phase

Sprint	Use Case Involved	Team members assigned
Sprint #1	US001 Login as User	Dheshighan A/L Saravana Moorthy
User module	US002 Register as Member	Evelyn Goh Yuan Qi
Sprint #2	US004 Apply for Savings Account	Chew Chiu Xian
Member module	US005 Making recurring payment	
	US006 Apply for Loan	Ng Yu Hin
Sprint #3	US003 Approve member application	Evelyn Goh Yuan Qi
Staff module		
Sprint #4	US007 View Report	Lau Yan Kai
Board of Director module	US008 Approve Loan Applications	Ng Yu Hin
	US009 Access Dashboard	Lau Yan Kai

2.4 User Story Details

2.4.1 US001: User Story <Login as User/>

Table 2.1: User Story Description for <Login as User>

User story: Login as user
ID: US001
User Story Description: As a KADA member and clerk I want to login as a member or admin of the cooperative So that I can access the benefits offered by the cooperative
Flow of events: 1.The user navigates to the login page. 2. The system prompts the user to enter their username and password. 3. The user enters their credentials and clicks the "Login" button. 4. The system validates the credentials. 5. If valid, the system sends a One-Time Password (OTP) to the user's registered WhatsApp number. 6. The user enters the OTP. 7. The system verifies the OTP. 8. Upon successful verification, the user is granted access to their dashboard.

Alternative flow:

1. If the user previously selected "Remember Me".
2. System pre-fills the username and email field.
3. User only needs to enter the password.
4. User enters their credentials and submits the form.
5. KADA system validates the credentials
6. KADA system grants access to the user and redirects them to the homepage.

Acceptance Criteria

Precondition:

- 1.The user must already be registered in the system.
- 2.The system must be operational.

Postcondition

- 1.The user is successfully logged in and granted access to the system.
- 2.The session is securely established for the user.

Exception flow:

1. If the username or password is invalid, the system displays an error message and allows the user to retry.
2. If the OTP is invalid, the system requests the user to re-enter or request a new OTP.

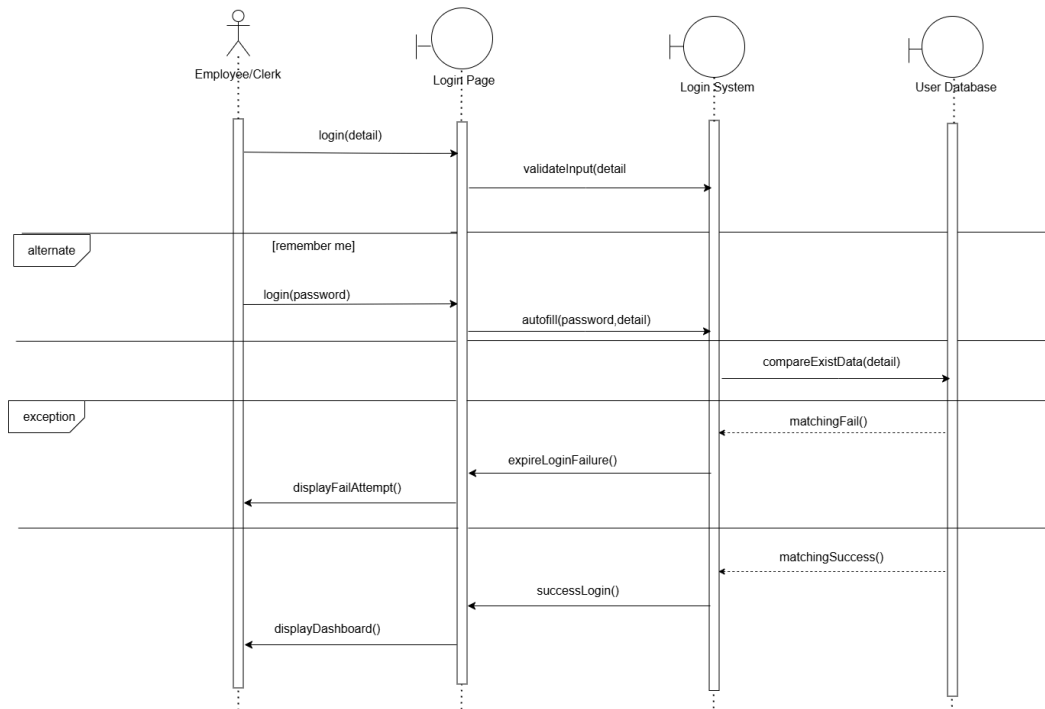


Figure 2.5: Sequence Diagram for <Name of User Story/Scenario if more than one scenario>

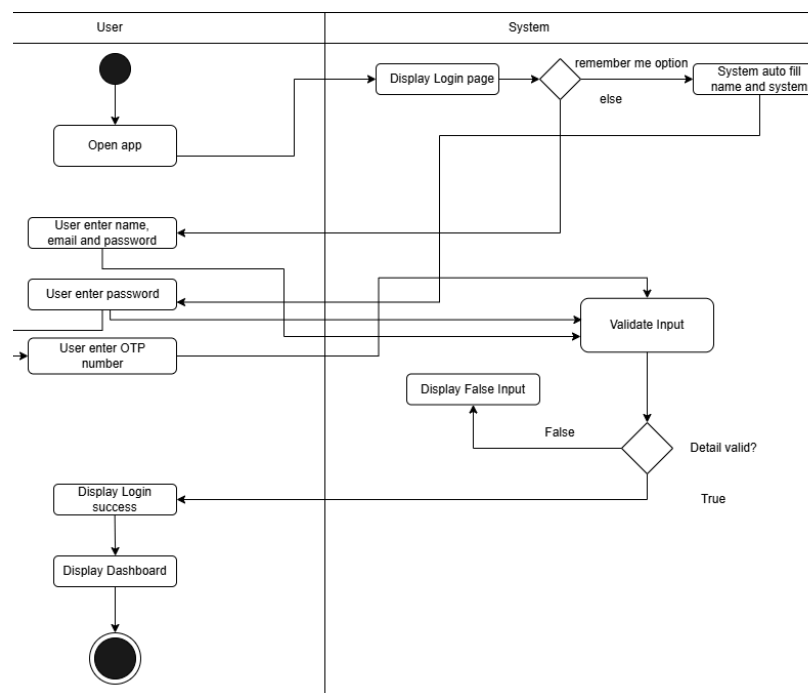


Figure 2.2: Activity Diagram for <<Name of User Story>>

2.4.2 US002: User Story <Register as member>

Table 2.1: User Story Description for <Register as Member>

User story: Register as member

ID: US002

User Story Description:

As a KADA employee

I want to register as a member of the cooperative

So that I can access the benefits offered by the cooperative

Flow of events:

1. The employee accesses the system.
2. The employee chooses “Membership Registration” on the navigation bar.
3. The employee fills in personal details and uploads required documents.
4. The employee clicks the submit button to submit the membership application form.
5. Confirmation message of the submission is displayed to the employee.

Alternative flow:

1. If mandatory fields are not filled, the system highlights missing fields to remind the employee to fill in the incomplete fields.
2. If document upload fails, the system prompts the employee to retry.

Acceptance Criteria

Preconditions:

1. The user must be a current employee of the KADA.
2. The employee must access the system and have a valid internet connection.

Postconditions:

1. If the application is approved, the employee will receive a notification.
2. If the application is rejected, the employee can reapply and will not be able to log into the system.
3. The employee's registration details are saved in the database, awaiting staff approval.

Exception flow:

1. System error during registration, prompting the employee to retry later.

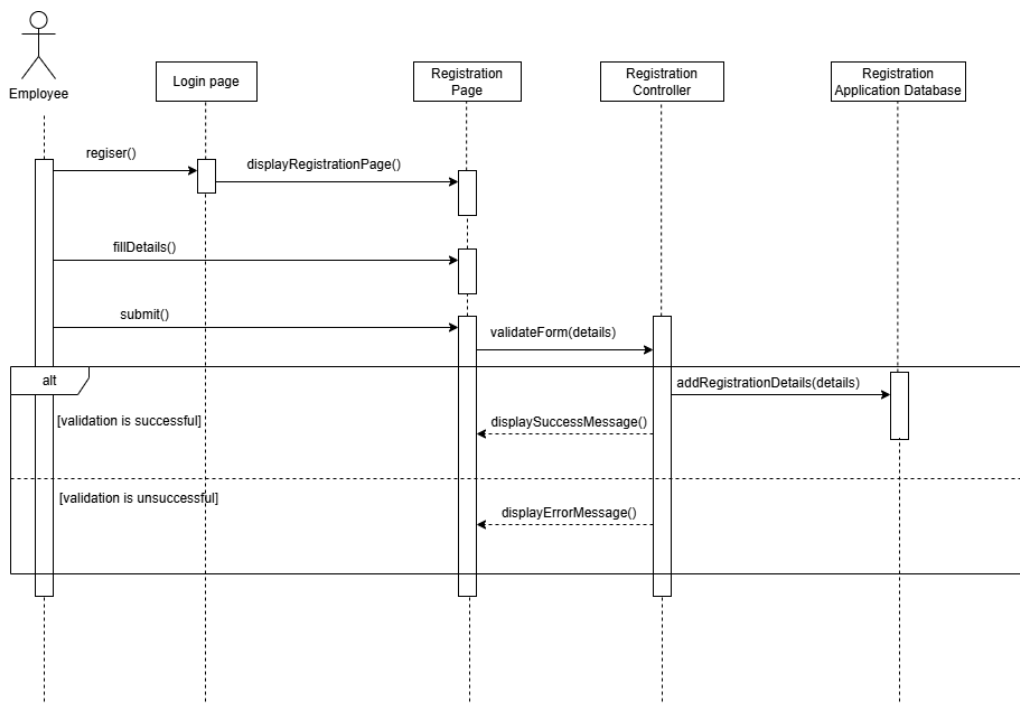


Figure 2.5: Sequence Diagram for <Register as member>

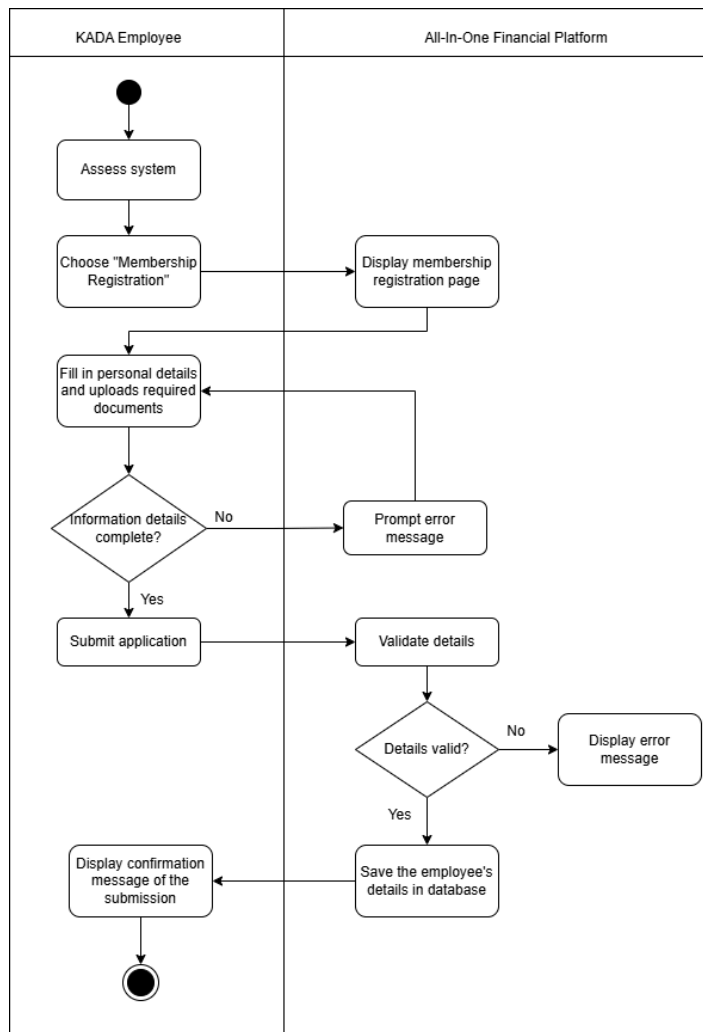


Figure 2.2: Activity Diagram for <<Register as member>>

2.4.3 US003: User Story <Approve member application>

Table 2.2: User Story Description for <Approve member application>

User story: Approve member application
ID: US003

User Story Description:

As a KADA staff

I want to approve new member application

So that I can determine whether the applicant meets the cooperative's membership requirements.

Flow of events:

1. Staff navigates to the member applications list.
2. Staff views pending member applications.
3. Staff reviews applicant's details and documents.
4. Staff selects either "Approve" or "Reject" the member applications.
5. System updates the status and notifies the applicants.

Alternative flow:

1. If approve the membership application, the system updates the application status to "Approved".
2. If reject the membership application, the system updates the application status to "Rejected".

Acceptance Criteria

Preconditions:

1. The system has pending member application requests.
2. Staff is logged in to the system

Postconditions:

1. The system updates the member application status.
2. Notifications are sent to the applicants informing them of the status.

Exception flow:

1. System error while updating the status. Staff is prompted to retry.

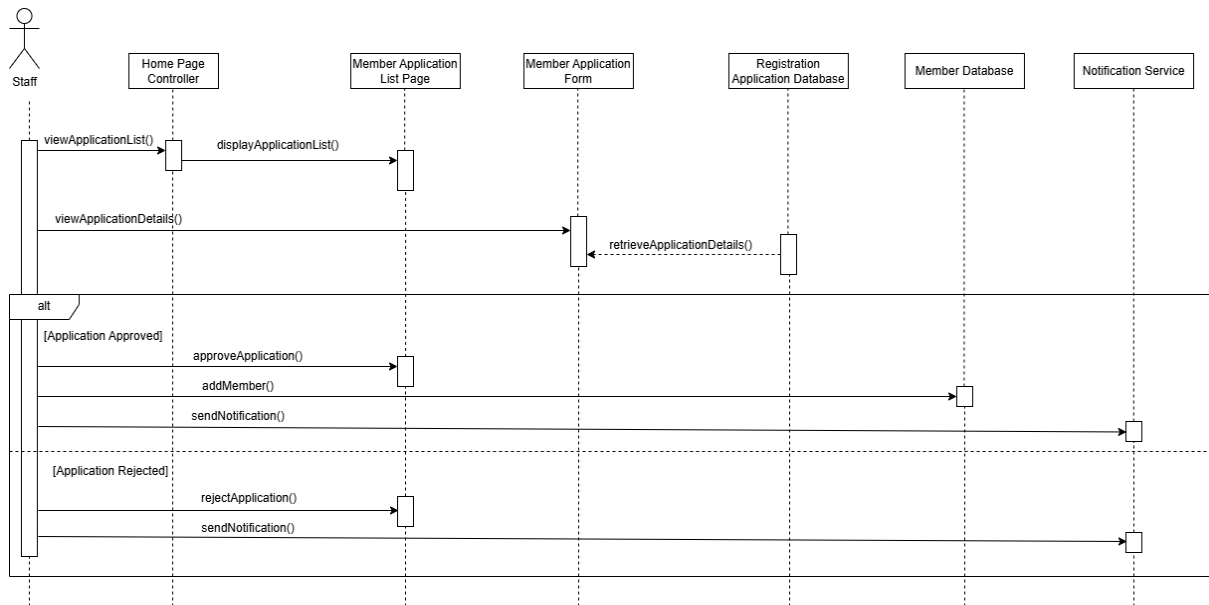


Figure 2.5: Sequence Diagram for <Approve member application>

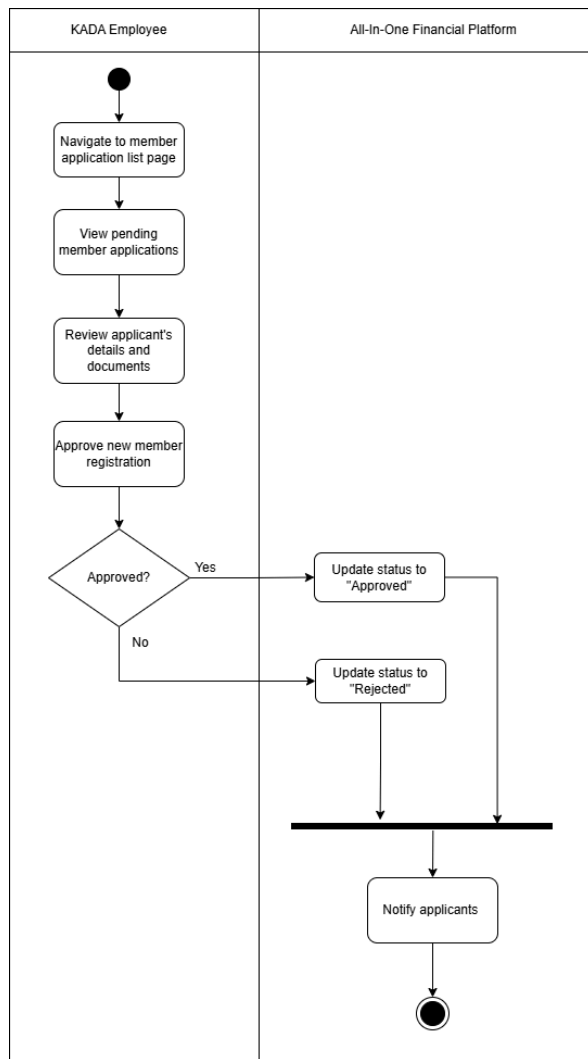


Figure 2.2: Activity Diagram for <<Approve member application>>

2.4.4 US004: User Story <Apply for Savings Account>

Table 2.3: User Story Description for <Apply for Savings Account>

User Story: Savings Account Application
ID: US004

User Story Description:

As a member of KADA,

I want to apply for a savings account and set savings goals,

So that I can securely save money and monitor my financial progress.

Flow of Events:

1. The user logs in into the system.
2. The user navigates to the "Savings" section of the platform.
3. The user selects "Apply for Savings Account."
4. The system prompts the user to enter account details such as target savings amount and duration.
5. The system validates the details and calculates a savings schedule.
6. The user reviews and confirms the application.
7. The system saves the savings account details in the database.
8. The user's dashboard is updated to display the new savings account and tracking tools.

Alternate Flow:

1. If mandatory fields are not filled, the system highlights missing fields to remind the employee to fill in the incomplete fields.

Acceptance Criteria:**Preconditions:**

1. The user must be a registered member of KADA.
2. The user must have logged into the system successfully.

Postconditions:

1. A new savings account is created and stored in the system.
2. The user's dashboard reflects updated savings progress.
3. The savings account appears on the user's dashboard with tracking tools.
4. The system validates the input and calculates a savings schedule.
5. Users can enter account details including target amount and duration.

Exception Flow:

1. If the user inputs invalid data (e.g., negative target amount), the system displays an error and prompts for correction.
2. If verification fails, the user is redirected to reattempt login.

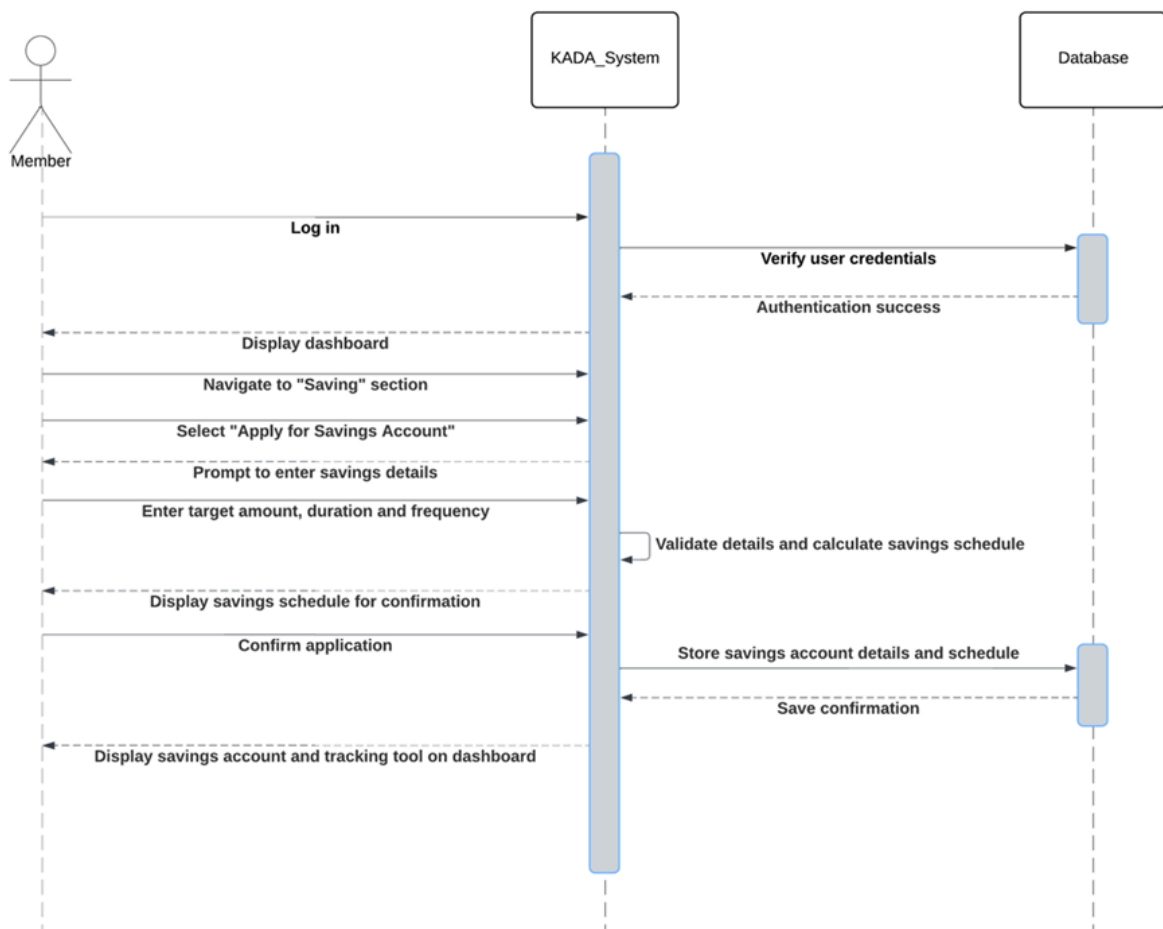


Figure 2.5: Sequence Diagram for <Apply for Savings Account>

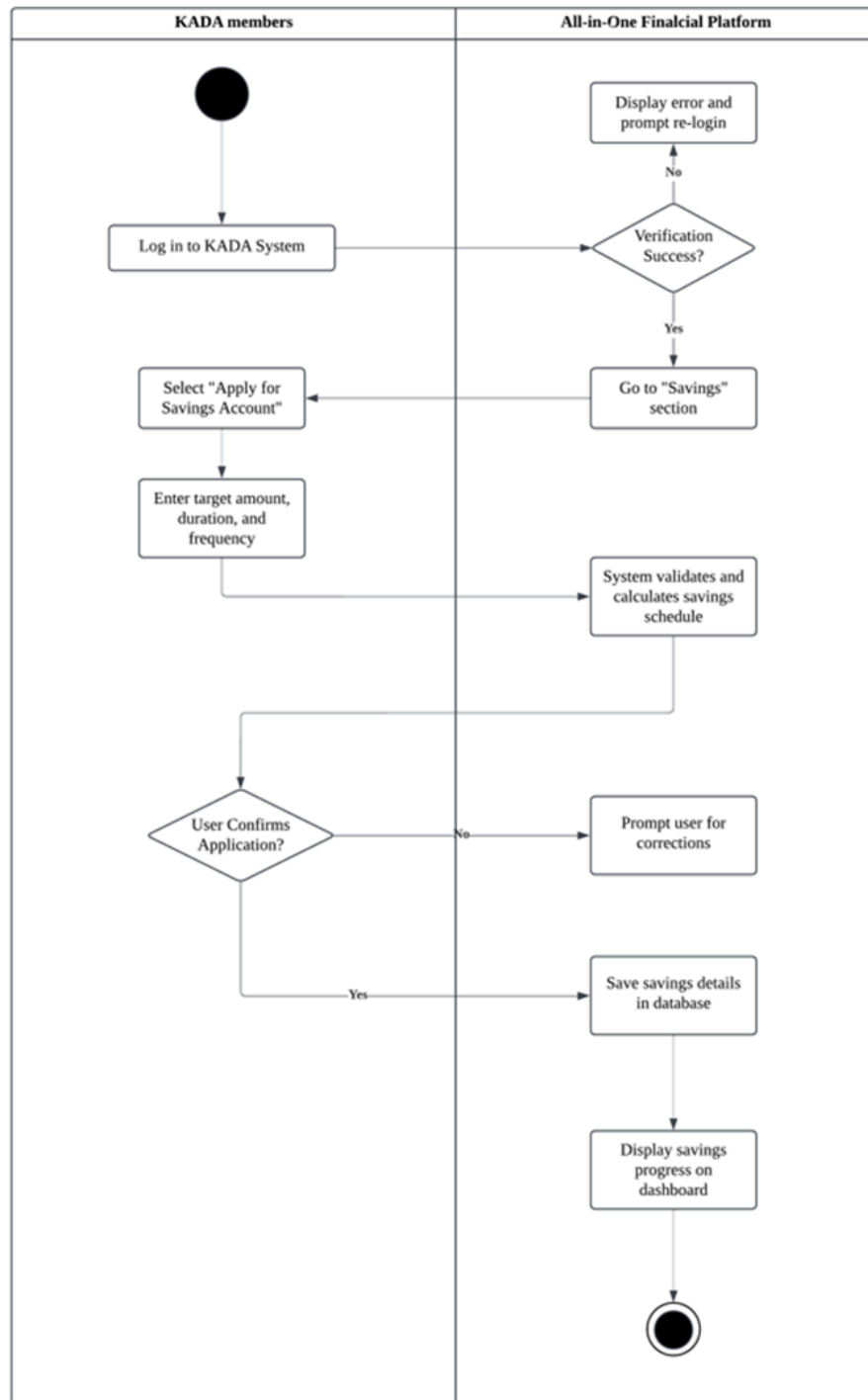


Figure 2.2: Activity Diagram for <<Apply for Savings Account>>

2.4.5 US005: User Story <Making recurring payment>

Table 2.4: User Story Description for <Making recurring payment>

User Story: Recurring Payment Option
ID: US005
User Story Description: As a member of KADA, I want to set up a recurring payment option for my savings account, So that I can automate deposits and achieve my savings goals consistently.
Flow of Events: 1. The user logs in into the system. 2. The user navigates to the "Savings" section of the platform. 3. The user selects "Recurring Payment Option." 4. The system prompts the user to enter payment details, such as frequency, amount, and payment method. 5. The system validates the payment details and sets up the recurring payment schedule. 6. The recurring payment details are saved in the database. 7. The system displays a confirmation message to the user. 8. The user's dashboard is updated with the recurring payment setup details.
Alternate Flow: 1. If mandatory fields are not filled, the system highlights missing fields to remind the employee to fill in the incomplete fields.

Acceptance Criteria:

Preconditions:

1. The user must be a registered member of KADA.
2. The user must have an active savings account.

Postconditions:

1. A recurring payment schedule is created and stored in the system.
2. The user's dashboard reflects the recurring payment details.
3. Users can enter payment details including frequency, amount, and method
4. The system validates the payment setup and confirms the recurring schedule.
5. Users view recurring payment details on their dashboard.

Exception Flow:

1. If the payment method is invalid or insufficient funds are detected, the system prompts the user to correct the information.
2. If the system cannot validate the payment setup, an error message is displayed.

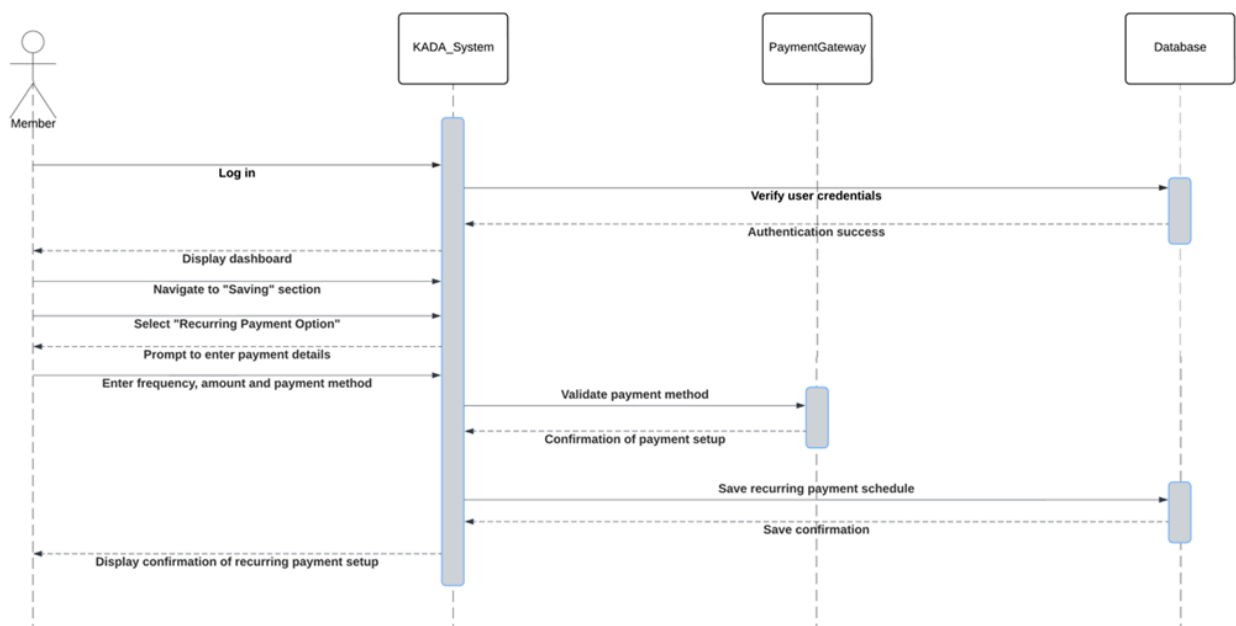


Figure 2.5: Sequence Diagram for <Making recurring payment>

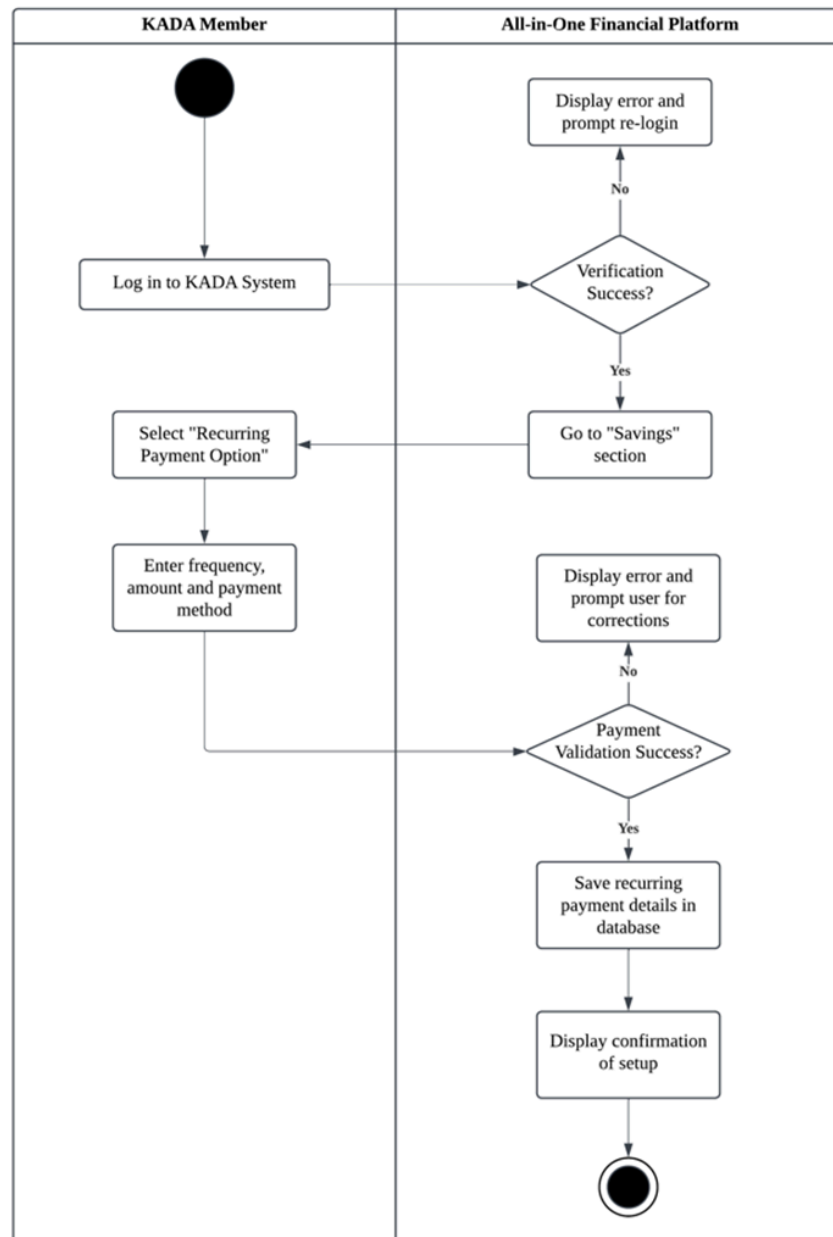


Figure 2.2: Activity Diagram for <<Making recurring payment>>

2.4.6 US006 User Story <Apply for Loan>

Table 2.5: User Story Description for <Apply for Loan>

User Story: Apply for Loan
ID: US006
User Story Description: As a member of KADA, I want to apply for requesting loans, So that to manage large expenditures by repaying the borrowed amount over time.
Flow of Events: 1. The member logs in into the system 2. The member navigates to the “Loans” section of the platform. 3. The member selects “Apply for Loans”. 4. The system displays a form with fields that are mandatory for the user to fill in with personal details and the amount wished to be requested. 5. The system validates the details. 6. The member reviews and confirms the application by clicking the submit button. 7. The system displays a confirmation message to the user. 8. The member awaits confirmation of loan status.
Alternate Flow: 1. If mandatory fields are not filled or incorrectly filled in, the system will highlight the fields to remind the member to fill in the incomplete fields or incorrect details.

Acceptance Criteria:**Preconditions:**

1. The member must be a registered member of KADA.
2. The member must have logged into the system successfully.

Postconditions:

1. If the application is approved, the member receives a notification.
2. If the application is rejected, the member can reapply.
3. The system saves the loan application form details in the database, awaiting staff approval.

Exception Flow:

1. If the user inputs invalid data (e.g., negative target amount), the system displays an error and prompts for correction.

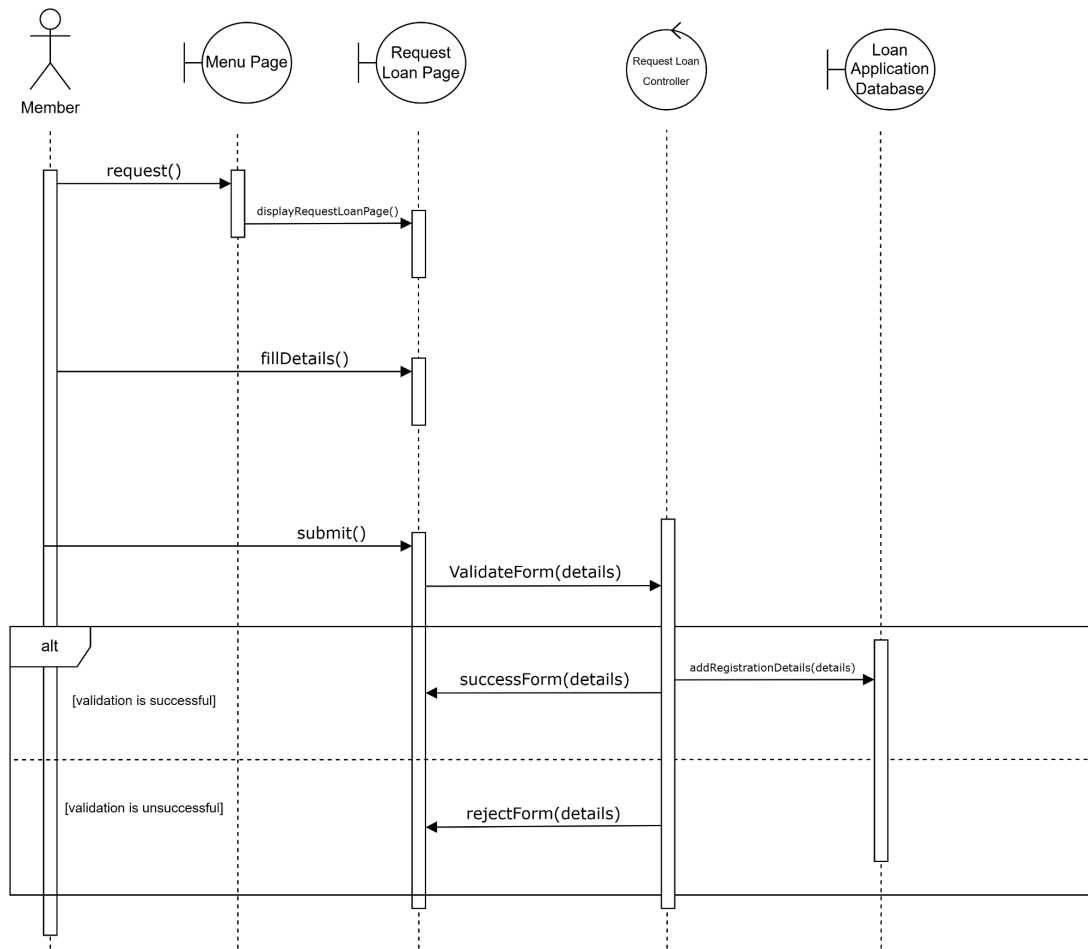


Figure : Sequence Diagram for <Apply for Loan>

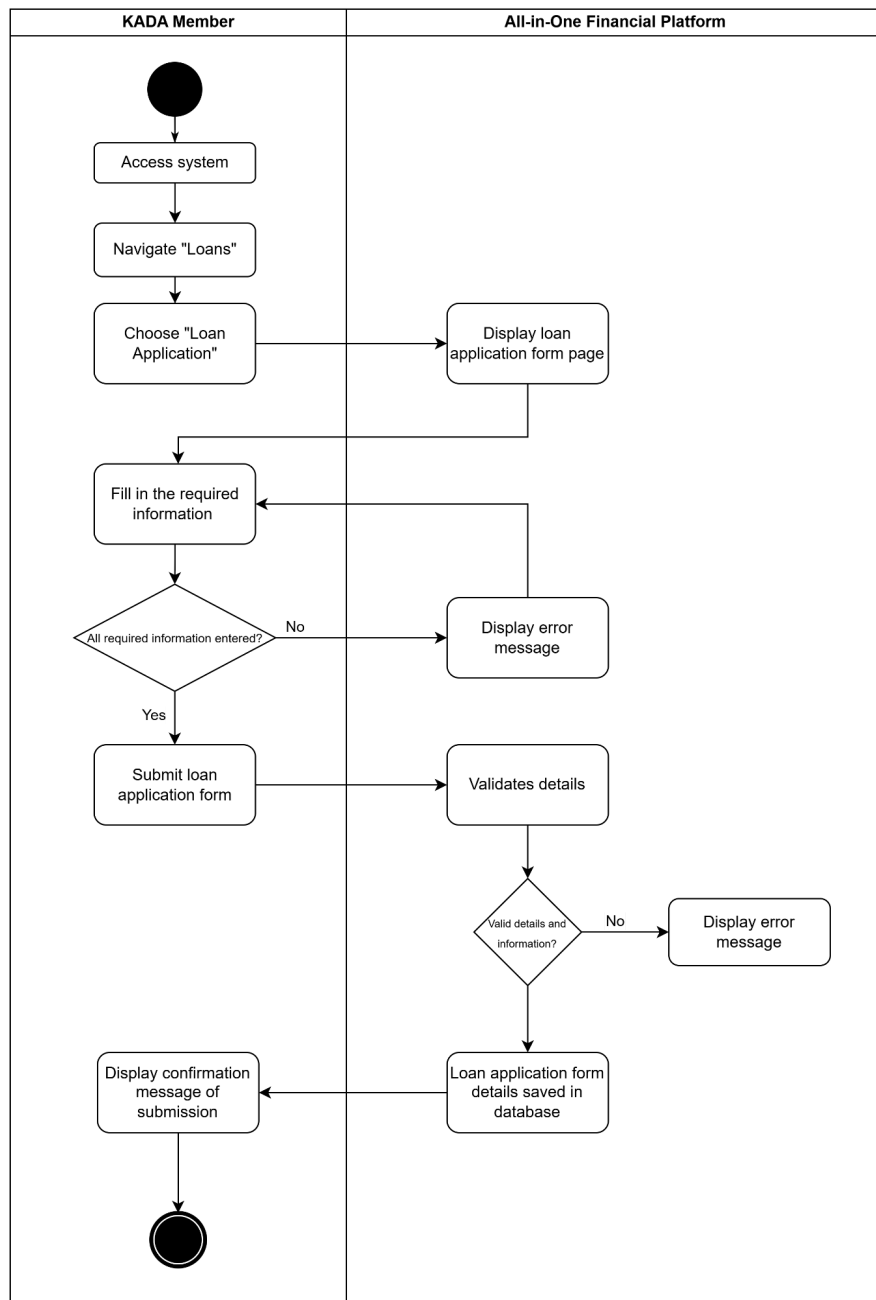


Figure 2.: Activity Diagram for <<Loan application >>

2.4.7 US007: User Story <View Report>

Table 2.2: User Story Description for <View Report>

User story: View Report
ID: US007
User Story Description: As a Director I want to view comprehensive financial report of the organisation So that I can evaluate the corporation's overall financial performance.
Flow of events: the director login into the system the director navigates to the "Reports" section the system displays a list of available financial reports, in monthly, quarterly, and yearly the director selects the desired report type and period the system retrieves and displays the selected financial report the director reviews the detailed breakdown of the report, with options to drill down into specific sections. the director can export the report as PDF, Excel file or print it directly the system displays a confirmation message when the report is successfully exported or printed.

Alternative flow:

If no financial reports are available for the selected period, the system displays a message that indicates no data is found.

The director can select a different period or adjust the filter criteria

Acceptance Criteria

Preconditions:

the director must have a valid account and login into the system

financial data must be preloaded and organised in the system

Postconditions:

the director can view the selected financial report

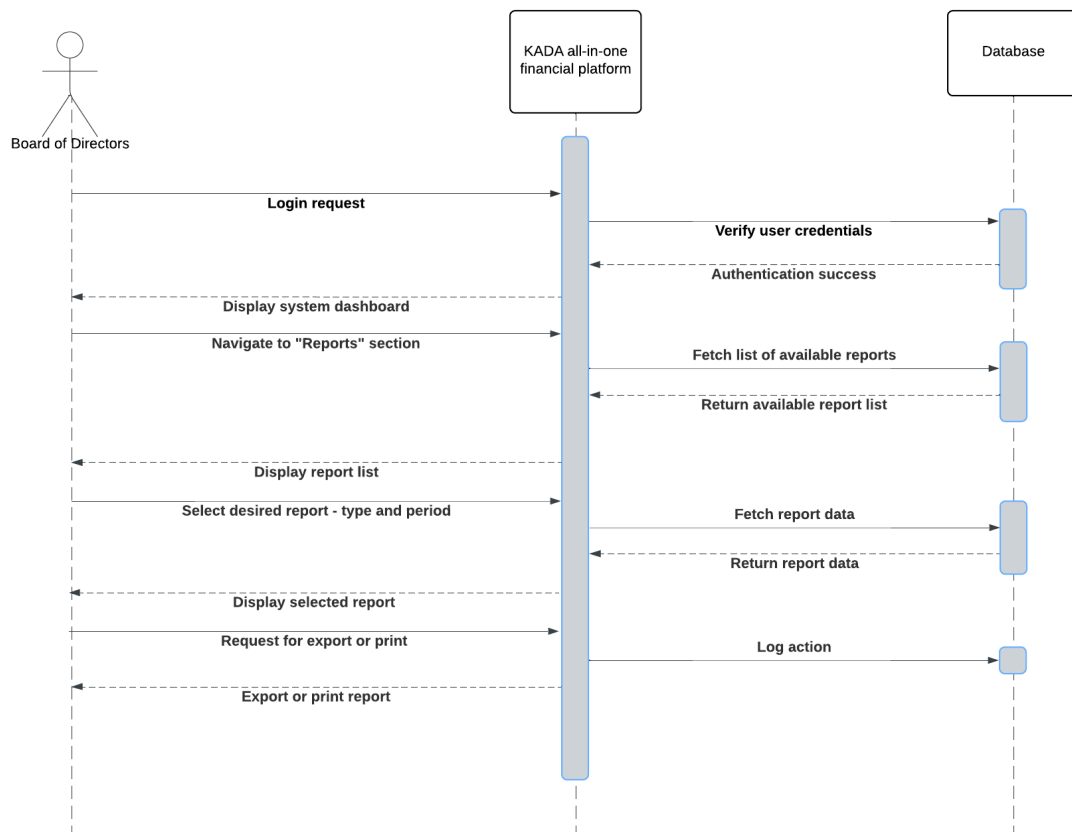
the system records the director's activity for audit purposes

exported or printed reports are correctly formatted and saved

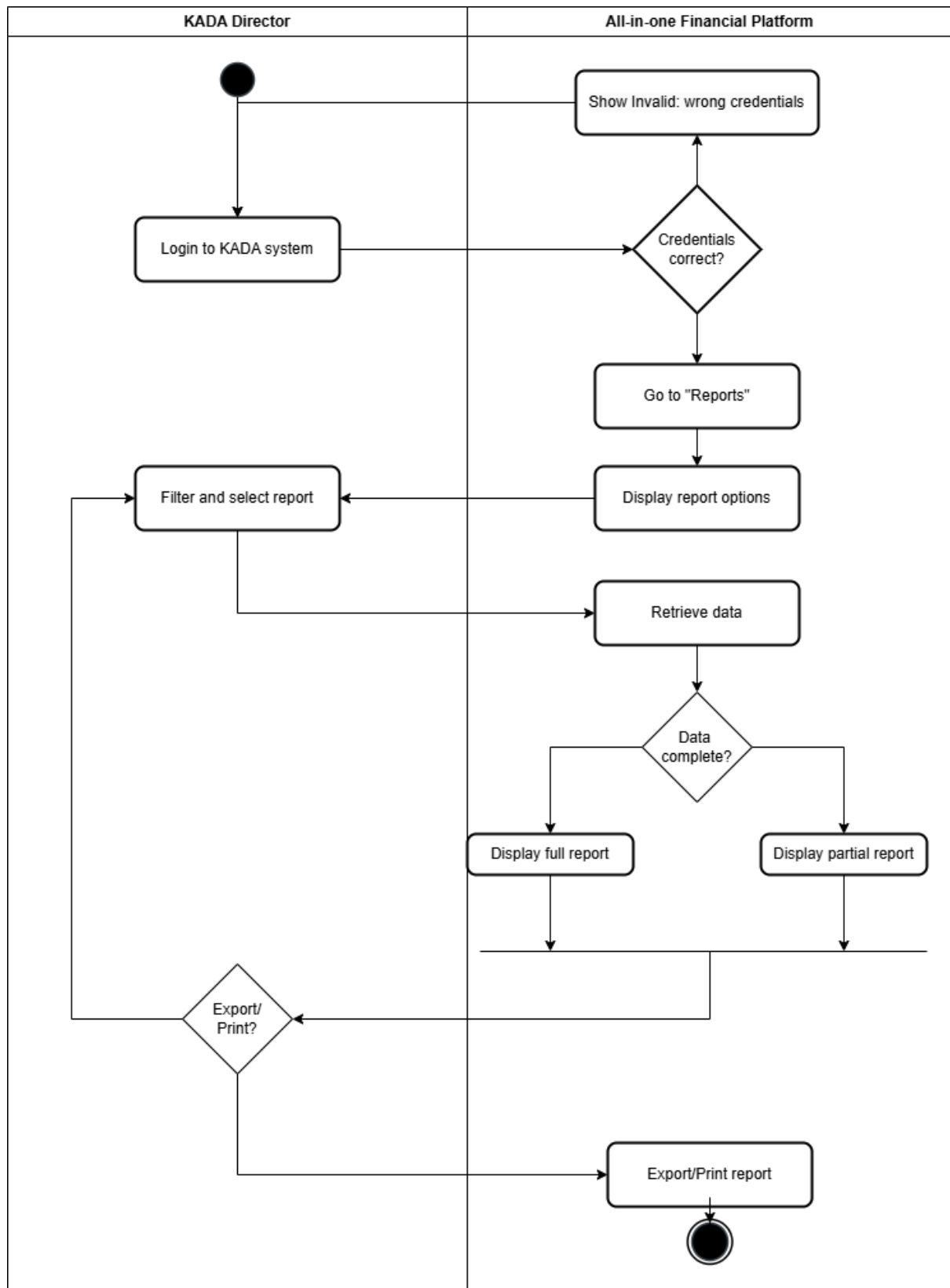
Exception flow:

if the system encounters error during data retrieval, it will display an error message and prompts the director to retry later

if there is a system outage, the director will be notified and the operation is logged for follow-up from the developers



Sequence <View Report>



Activity <View Report>

Table 2.2: User Story Description for <Approve Loan Applications>

User story: Approve Loan Applications
ID: US008
User Story Description: As a Director I want to review and approve large amount of loan applications So that I can make sure the corporation is operating within financial limit
Flow of events: The director login into the system the director navigates to the “Loan Applications” section. the system displays a list of all pending loan applications that require director approval the director selects a specific loan application to review the system displays detailed loan information, such as the applicant details, loan amount, supporting documents, purpose of loan, risk analysis and financial impact assessment the director reviews the details and may add comment or request additional documentation if needed the director choose to either “Approve” or “Reject” the loan application. the system prompts the director to confirm the decision upon confirmation, if approved, the system update the loan status to “Approved” and notifies the applicant if rejected, the system updates the loan status to “Rejected” and notifies the applicants the system logs the decision for audit purpose.

Alternative flow:

if the director did not decide immediately, the application remains in the pending list.

Acceptance Criteria**Preconditions:**

the director must have a valid account to login into the system

loan applications that require director approval must exist in the system

supporting financial and risk analysis data must be accessible

Postconditions:

approved loan applications are updated in the system, and the applicants are notified

rejected loan applications are updated in the system, and the applicants are informed.

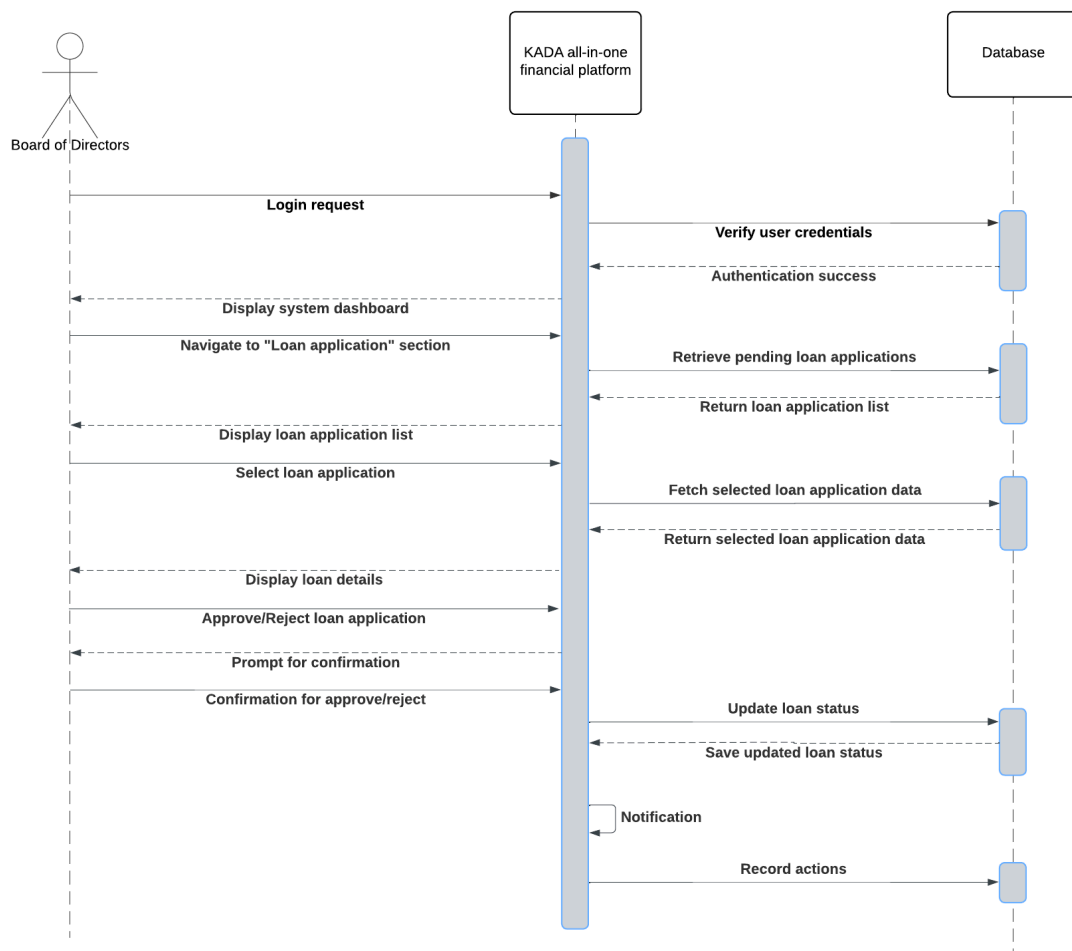
all actions are logged for audit purposes.

Exception flow:

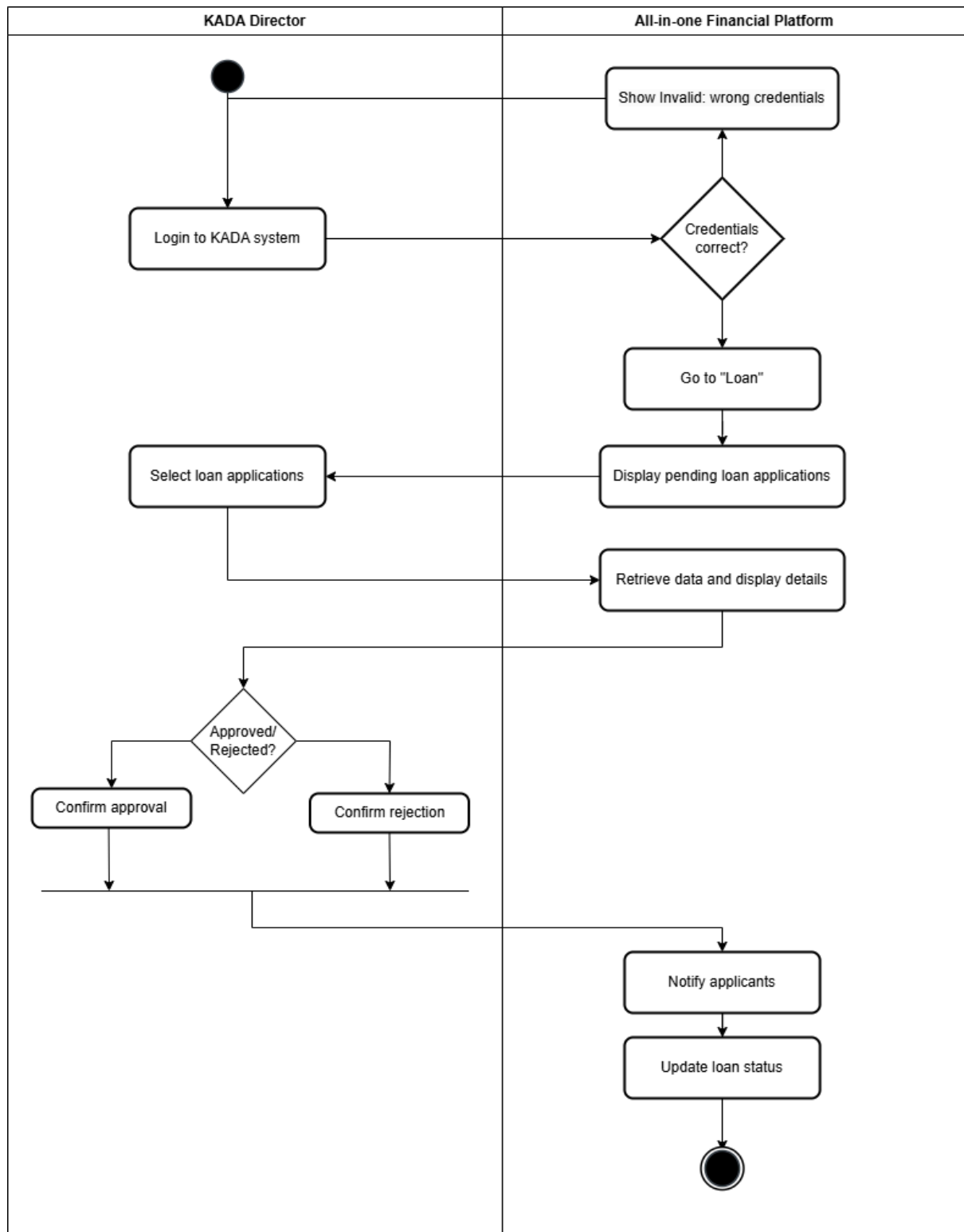
if the system encounters an error during the review process, it will display an error message.

if the director's session expires or the system goes offline, the application remains pending for review later.

if the loan exceeds allowable limits, the system alerts the director and provides an option to escalate or revise.



Sequence diagram <Approve Loan Applications>



Activity <Loan Approval>

Table 2.2: User Story Description for <Access Dashboard>

User story: Access Dashboard

ID: US009

User Story Description:

As a Director

I want to access a dashboard that summarise key metrics and activities

So that I can make informed decisions

Flow of events:

The director login into the system

the director is automatically redirected to Dashboard section

the system loads and displays a dashboard with key metrics, such as financial performance indicators, operational KPIs, alerts and notifications, graphical summaries

the director clicks on specific metrics or sections for a more detailed breakdown

the director can customise the dashboard layout by adding, removing, or rearranging the widgets

the director can export a report summarising the dashboard data or share it with stakeholders via email

the system logs the director's activities for audit purposes

Alternative flow:

if the dashboard fails to load due to network or system issue, the system displays a retry option or offline summary

if data is incomplete for certain metrics, the system highlights them and provide an explanation

Acceptance Criteria**Preconditions:**

the director must have a valid account to access the data

key data points must be preloaded and updated in system

Postconditions:

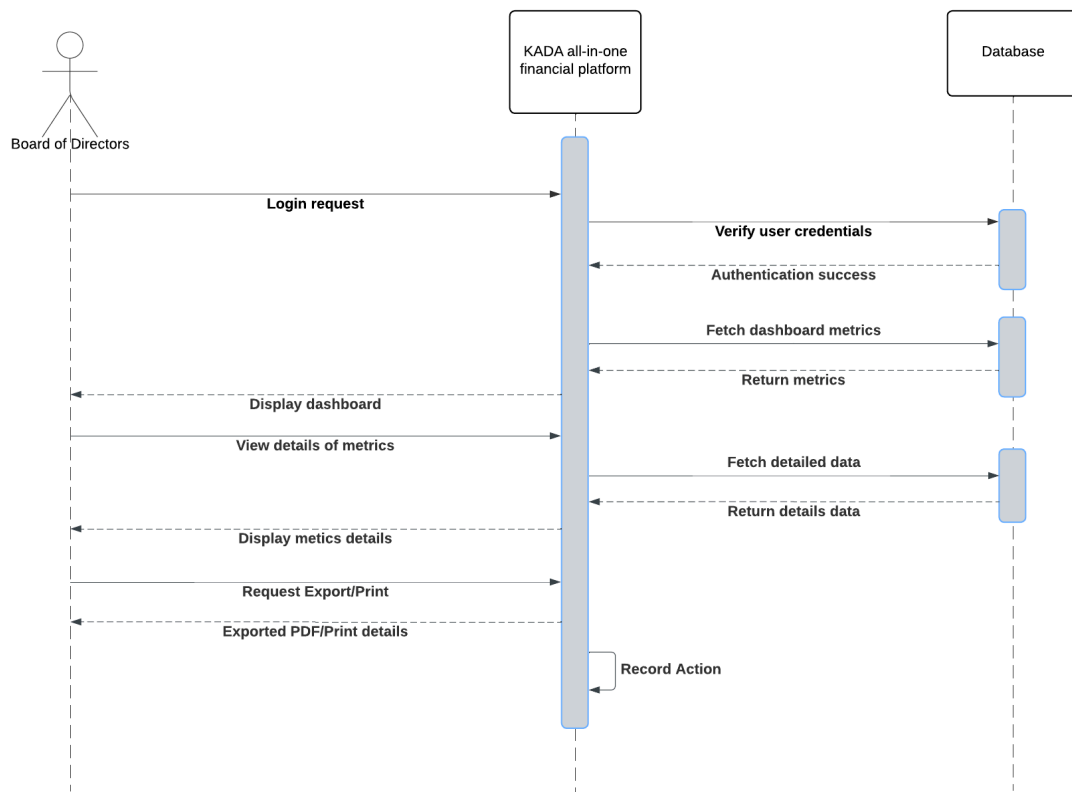
the director gains a clear overview of the corporation performance and activities

exported or shared dashboard summaries are correctly formatted and include up-to-date date information

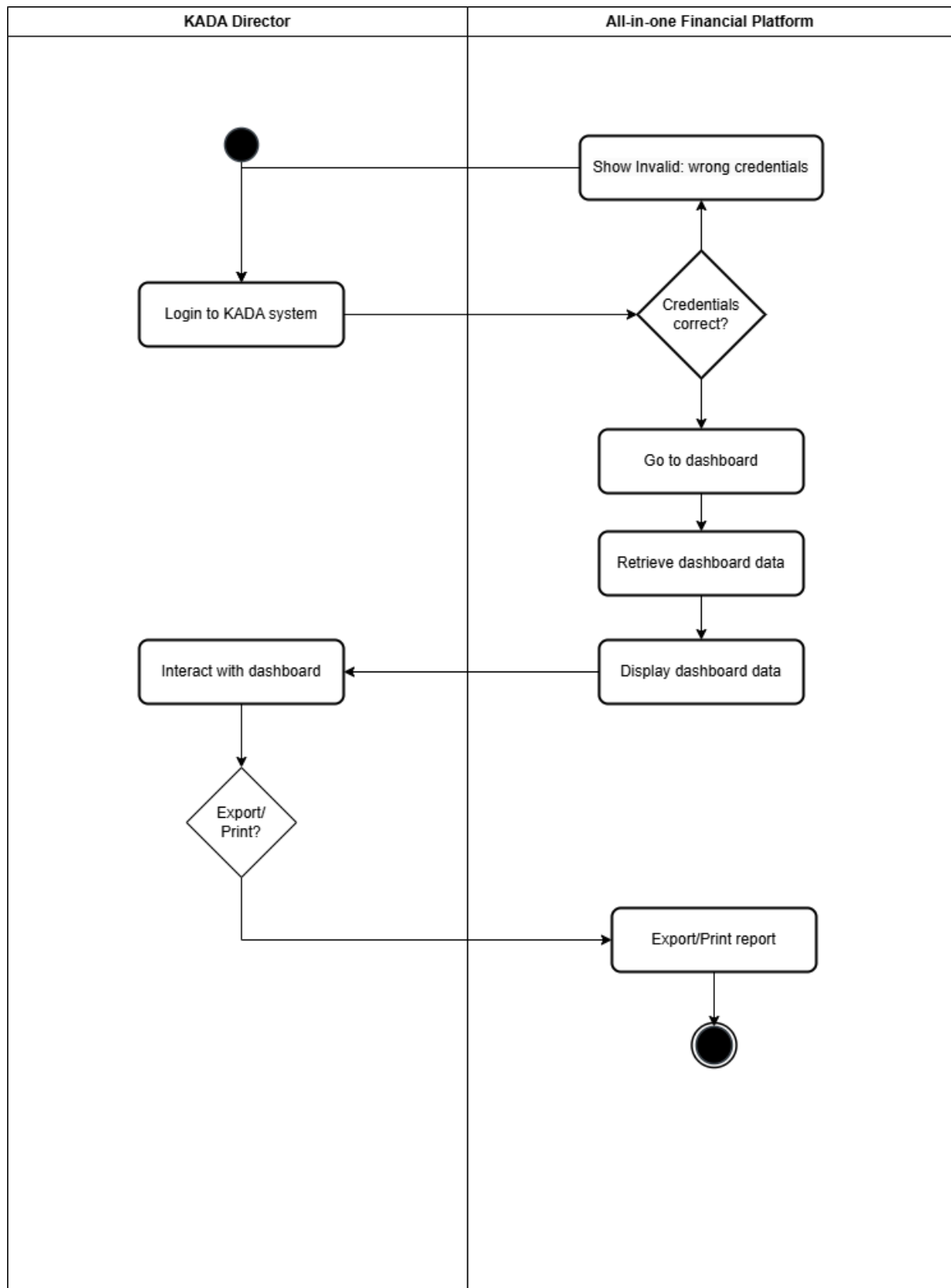
Exception flow:

if the system encountered an error, it will display an error message and log issue for resolution

if the director's session times out, the system prompts for re-authentication and reload the dashboard upon login



Sequence <View Dashboard>



Activity <View Dashboard>

2.5 Performance and Other Requirements

Non-functional requirements refer to how the Koperasi KADA all-in-one financial platform system should behave to ensure a high-quality user experience. These requirements are categorised into **Performance Requirements**, **Software System Attributes**, and **Other Requirements**.

2.5.1 Performance Requirements

I. Response Time

The system should respond quickly to user actions, like logging in or retrieving data. Tasks such as uploading documents or submitting loan applications should not take too long.

II. Throughput

The platform should handle many user requests at the same time without slowing down. It should process multiple transactions efficiently, even during busy times.

III. Capacity

The system should support many users and store all necessary data without performance issues. It should retain records for several years to meet operational needs.

IV. Availability

The system should be available most of the time, with minimal interruptions. Planned maintenance should not affect users significantly, and users should be informed in advance.

2.5.2 Software System Attributes

I. Usability

The platform should be easy to use, even for users with limited technical skills. Accessibility features such as support for screen readers and multiple languages should be included in the system.

II. Reliability

The system should work consistently without unexpected errors. If there is a failure, it should recover quickly to avoid disruptions.

III. Portability

The system should work well on different devices, including computers and smartphones. It should also run smoothly on various browsers and operating systems.

IV. Security

User data should be protected through encryption and secure login methods. The system should be designed to prevent unauthorised access and other security threats.

2.5.3 Other Requirements

I. Maintainability

The system should be easy to update and fix when needed. Adding new features or resolving issues should not require too much time or effort.

II. Compatibility

The platform should connect easily with external systems like accounting tools and APIs. It should allow users to export data in common formats, such as CSV or PDF.

III. Legal and Regulatory

The system should comply with relevant laws, regulations, and standards regarding data privacy, confidentiality, and information security.

IV. Environmental Impact

The system should be designed to use resources efficiently and reduce energy consumption.

2.6 Design Constraints

Koperasi KADA All-in-one Financial Platform is bound by several design constraints to guarantee its functionality, usability, and compliance within the intended environment.

Environmental constraints:

The system is utilized by KADA employees, including users, members, and administrators, from various devices and locations. Therefore, it must perform reliably under common indoor conditions, such as office environments and hostels, with standard temperature and lighting. For instance, the system should function effectively in environments with temperatures ranging from 10°C to 35°C and moderate humidity levels.

Hardware constraints:

The system must perform efficiently on devices with a minimum of 4GB of RAM to support the diverse hardware typically used by KADA employees and administrators. Furthermore, it should be compatible with both desktop and mobile platforms, including Android and iOS operating systems.

Security constraints:

The system must adhere to Malaysia's Personal Data Protection Act 2010 (PDPA), ensuring the secure handling of personal data, including its collection, storage, and processing. It should employ AES-256 encryption for robust data protection and implement measures such as secure authentication, access controls, and encrypted communication channels. Moreover, the system must comply with cybersecurity standards set by CyberSecurity Malaysia to maintain consistent data security and privacy compliance.

Compatibility constraints:

The system must seamlessly integrate with existing campus systems, including student portals and academic databases. It must also ensure compatibility with popular browsers such as Chrome, Firefox, Safari, and Edge. Additionally, the system should function effectively on both older and newer operating systems, including Windows 10/11, macOS 10.15 and later, Android 8.0 and above, as well as iOS 13 and above.

Performance constraints:

The system must be designed to support up to 1,000 concurrent users, with a maximum response time of 3 seconds per request. Furthermore, it should be capable of dynamically scaling to handle increased demand during peak usage periods, such as exam times, when mental health resources are more heavily utilized.